

# Matthew Stuart Jones

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## EDUCATION

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### University of Kentucky

Bachelor of Science, Computer Science

Projected Graduation, May 2027

GPA: 4.0/4.0

- Minors: Mathematics and Vocal Performance
- Artificial Intelligence Certificate
- Study Abroad Experience in Bilbao, Spain (Universidad de Deusto)

## RESEARCH EXPERIENCE

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### NSF Research Experience for Undergraduates (REU), AI for Social Good

Summer 2026

Worcester Polytechnic Institute (WPI) — Worcester, Massachusetts

- Selected for nationally competitive NSF-funded REU cohort in Data Science and Artificial Intelligence
- Research focused on adverse drug reactions (ADRs), mapping interconnections between drug-drug, drug-herb, and drug-supplement interactions using ML and deep learning methods
- Developing adaptable modeling approaches across diverse biomedical problem domains

### Independent Study: Responsible AI Citation & Policy Influence Mapping

Spring 2026

Advisor: Dr. Brent Harrison, University of Kentucky — Human Centered AI, AI Ethics

- Analyzed citation flows between RAI literature (AIES, NeurIPS) and 18 national AI governance frameworks; identified three-way misalignment between paper production (fairness), citation impact (explainability), and policy language (accountability)
- Built end-to-end data pipeline using Semantic Scholar API, intfloat/e5-large-v2 embeddings, and cosine similarity scoring to quantify research-to-policy alignment across a 20,000+ paper corpus
- Presented findings at the 20th Annual University of Kentucky Showcase of Undergraduate Scholars

### AI Researcher for ACCESS

October 2024 – Present (incl. Summer 2025)

UK Center for Computational Sciences (CCS) — Lexington, Kentucky

- Built MCP-based semantic router to categorize incoming queries and route them to the appropriate AI agent across the ACCESS cyberinfrastructure ecosystem
- Developed Retrieval-Augmented Generation (RAG) system to assist users in drafting allocation abstracts and proposals based on a corpus of past submissions
- Designed multi-LLM tournament tagger to categorize research proposals by field of science using open-source models deployed on a Grace Hopper cluster via Ollama
- Improved existing AI models through document verification and retraining workflows

## INVITED TALKS

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### Guest Lecturer, HON 251: Technology and the Human Experience (Honors)

Fall 2025

“A Brief Overview of AI Policy” — University of Kentucky

## CONFERENCES

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- “UK NSF Regional AI Workshop” — Lexington, Kentucky October 7, 2025
- “UK AI/ML Research Symposium” — Lexington, Kentucky March 28, 2024; November 12, 2024
- “UK Commonwealth Computational Sciences Summit on AI” — Lexington, Kentucky October 17, 2023

## SKILLS

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- **AI/ML Engineering:** Built RAG systems, MCP semantic routers, and LLM-based tagging pipelines using LlamaIndex, ChromaDB, sentence-transformers, and Ollama; experience with OpenAI API and local open-source models.
- **Mathematical Foundations:** Strong background in algorithms, linear algebra, probability, and numerical methods; coursework in sequential decision making, deep learning, and models of computation.
- **Programming & Systems:** Python (NumPy, pandas, scikit-learn) for ML/AI; C and C++; SQL; full-stack web development (JavaScript, HTML, CSS); Linux; R.
- **Research & Data Pipelines:** Designed end-to-end NLP pipelines for literature analysis at scale — API-based corpus ingestion, embedding generation (e5-large-v2), and cosine similarity scoring across large document corpora; familiar with academic literature APIs (Semantic Scholar, OpenAlex).

- **Policy Communication:** Translates complex AI and technical concepts for policy, academic, and general audiences; experience analyzing national AI governance frameworks across multiple jurisdictions; public speaker and technical presenter; fluent in Spanish.

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## FEATURED COURSEWORK

|                |                                     |                |                             |
|----------------|-------------------------------------|----------------|-----------------------------|
| <b>CS 667</b>  | Sequential Decision Making          | <b>CS 463G</b> | Intro to AI                 |
| <b>CS 575</b>  | Models of Computation               | <b>CS 315</b>  | Algorithms                  |
| <b>CS 485G</b> | Data Ethics                         | <b>PPL 306</b> | Ethics and Civic Leadership |
| <b>MA 322</b>  | Matrix Algebra and Its Applications | <b>MA 321</b>  | Intro to Numerical Methods  |
| <b>MA 320</b>  | Introductory Probability            |                |                             |

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## HONORS AND AWARDS

- Dean's List — 6 Semesters
- Tau Beta Pi Scholar
- Pigman College of Engineering Marky Campbell Scholarship
- UK Bluegrass Spirit Scholarship
- UK Education Abroad Scholarship

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## PROFESSIONAL ORGANIZATIONS

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| <b>Kentucky Alpha Chapter of Tau Beta Pi</b>                                                                                                                                                                                                                                                                                      | Tau Beta Pi Scholar & Chapter Treasurer |
| <ul style="list-style-type: none"> <li>• Manage chapter finances for a 120-member engineering honor society, including dues, budgeting, and scholarship requests</li> <li>• Coordinate food and logistics for chapter meetings and events</li> <li>• Volunteer judge at Future Cities regional engineering competition</li> </ul> |                                         |

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## ORGANIZATIONS AND INVOLVEMENT

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| <b>acoUstiKats A Cappella</b>                                                                                                                                                                                                                                                                                                                            | Social Media Manager    |
| <ul style="list-style-type: none"> <li>• Spearheaded the group's first ICCA campaign where I initiated the decision to compete and led competition research</li> <li>• Create content across platforms, design and produce group merchandise, and publicize campaign events</li> </ul>                                                                   |                         |
| <b>UK Men's Chorus</b>                                                                                                                                                                                                                                                                                                                                   | Baritone Section Leader |
| <ul style="list-style-type: none"> <li>• Served as section leader on the group's New York tour, coordinating logistics and welfare for the baritone section</li> <li>• Part of a leadership team focused as much on culture as performance, fostering an environment where singers are locked in and also genuinely taking care of each other</li> </ul> |                         |

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## REFERENCES

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| <b>Dr. Brent Harrison</b><br>Director of Undergraduate Studies, CS<br>Pigman College of Engineering, UK<br>brent.harrison@uky.edu<br>(859) 257-8040 | <b>Dr. Jim Griffieon</b><br>Director, UK Center for Computational Sciences<br>University of Kentucky<br>griff@netlab.uky.edu<br>(859) 257-6746 |
| <b>Dr. Mike Johnson</b><br>Tau Beta Pi Faculty Advisor<br>University of Kentucky<br>mike.johnson@uky.edu<br>(859) 257-0717                          | <b>Dr. Cory Curl</b><br>Director of Undergraduate Studies<br>Martin School of Public Policy, UK<br>cory.curl@uky.edu<br>(859) 562-3182         |
| <b>Dr. Ruiquan Huang</b><br>Assistant Professor, Department of CS<br>University of Kentucky<br>ruiquanhuang@uky.edu<br>(859) 257-3391               |                                                                                                                                                |